

Abstract for ECOC 2026

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Quantum Advantage with Continuous Variable Systems

Abstract

Continuous-variable (CV) optical quantum systems are achieving remarkable milestones in quantum information science and technology. Recent breakthroughs include the demonstration of Gaussian boson sampling, the use of squeezed states for extreme quantum sensing in gravitational-wave detection, and significant advances in continuous-variable quantum key distribution (CVQKD) for secure quantum communication. Together, these developments highlight the versatility of CV systems and lay the groundwork for future experiments and applications.

In this talk, I will present our efforts to leverage CV optical quantum systems for achieving quantum advantage in learning, communication, and computation. Highlights include demonstrations of massive quantum advantage in quantum learning, progress toward quantum computational advantage, and recent advances in practical CVQKD. I will discuss these developments, address key challenges, and provide an outlook for the future of CV quantum information science.